

Campbell–Rényi under level prices: verification, repair, and a Lean ladder

Draft (2026-08-09). Desk report for the campbell-renyi track. This document makes no design decision; it verifies the proposed correspondence and records the choices exposed by making it exact.

Claim labels used throughout:

- **Verified** means checked against a cited primary source or against an already mechanized fact in this repository.
- **Desk-proved** means a complete finite derivation is given here but is not yet mechanized.
- **Conjectured** means a plausible extension whose proof is not given.
- **Wall** means the stated target cannot honestly be obtained without changing the representation or adding a hypothesis.

1. Verdict first

Desk-proved. The order claimed in shannon-programme.md is right: for

$$c_t(\ell) = 2^{t\ell}, \quad t > 0, \quad \alpha = \frac{1}{1+t},$$

the continuously optimal mount shares are the escort distribution

$$r_i = \frac{p_i^\alpha}{Z_\alpha}, \quad Z_\alpha = \sum_j p_j^\alpha,$$

and hence

$$d_i^* = -\log_2 r_i = \log_2 Z_\alpha - \alpha \log_2 p_i.$$

Thus the informal slope $d_i \approx \alpha \log_2(1/p_i)$ is correct, but it omitted the common normalizing offset $\log_2 Z_\alpha$. That offset is not optional: it is what makes Kraft tight.

Desk-proved, correction. The optimal *value* needs more careful wording. Campbell minimizes the logarithmically normalized exponential moment

$$L_t(d) = \frac{1}{t} \log_2 \sum_i p_i 2^{td_i}.$$

Our machine instead charges cumulative travel

$$C_t(d) = \sum_{\ell=0}^d 2^{t\ell}.$$

For integer d this is an affine function of 2^{td} , so it has the same optimizer, but its optimum is an affine function of 2^{tH_α} , not H_α itself. The surviving precise slogan is:

Exponential level prices select a Rényi order; normalized log travel is Rényi, while raw travel is exponential in Rényi.

Desk-proved, correction. The mounting constraint does not change: Kraft remains the antichain-packing law. Only the objective changes. “Level prices turn Kraft into Rényi” is therefore half-right. The correct version is:

Kraft stays Kraft; exponential travel turns Shannon’s objective into Campbell’s objective, whose optimum is Rényi.

Wall (representation, not mathematics). For $t = 1/2$ and $t = 1/3$, the literal prices $2^{t\ell}$ are generally irrational. An exact natural-valued Lean cost cannot at the same time be literally that schedule. There are three honest choices: use exact algebraic/real costs; round each level price to a natural; or group k binary levels into one natural-priced macro-level. Section 4 recommends the last as the first all- $\mathbb{N}$ theorem layer and a sandwich theorem relating it to the literal model. There is no honest “exact $\mathbb{N}$ form” of an irrational equality without making such a representation choice.

2. Continuous check from scratch

Fix a finite probability vector $p_i > 0$ with $\sum_i p_i = 1$. Heads with $p_i = 0$ can be removed from the optimization; in the finite machine they may later be placed in unused deep slots. Put $q = 2^t > 1$.

2.1 Cumulative travel reduces to an exponential moment

Desk-proved. With the convention that distance zero still pays the level-zero touch, matching Expo 2’s $1 + d$ when $t = 0$,

$$C_t(d) = \sum_{\ell=0}^d q^\ell = \frac{q^{d+1} - 1}{q - 1} = \frac{q}{q - 1} q^d - \frac{1}{q - 1}.$$

Therefore

$$T_t(p, d) := \sum_i p_i C_t(d_i) = \frac{q}{q - 1} M_t(p, d) - \frac{1}{q - 1}, \quad M_t(p, d) := \sum_i p_i 2^{td_i}.$$

The affine coefficient is positive, so T_t and M_t have exactly the same minimizers. This is where machine travel and Campbell meet. They differ only after the minimizer has been chosen: Campbell reports $L_t = (1/t) \log_2 M_t$, while the machine reports T_t .

2.2 Lagrange calculation

Desk-proved. Relax d_i from naturals to nonnegative reals and minimize M_t under $\sum_i 2^{-d_i} \leq 1$. Except for the one-head degenerate case, the constraint is tight: if it had slack, every d_i could be decreased slightly and the objective would fall. With multiplier λ ,

$$\mathcal{L}(d, \lambda) = \sum_i p_i 2^{td_i} + \lambda \left(\sum_i 2^{-d_i} - 1 \right).$$

Stationarity gives

$$tp_i 2^{td_i} = \lambda 2^{-d_i}, \quad 2^{(1+t)d_i} = \frac{\lambda}{tp_i}.$$

Thus 2^{-d_i} is proportional to $p_i^{1/(1+t)}$. Writing $\alpha = 1/(1+t)$ and normalizing by Kraft yields

$$2^{-d_i^*} = r_i = \frac{p_i^\alpha}{Z_\alpha}, \quad d_i^* = \log_2 Z_\alpha - \alpha \log_2 p_i.$$

Every r_i lies in $(0, 1]$, so the nonnegativity constraints on the d_i introduce no extra case.

2.3 Optimal value

Desk-proved. Since $\alpha t = 1 - \alpha$,

$$\begin{aligned} M_t(p, d^*) &= \sum_i p_i r_i^{-t} \\ &= Z_\alpha^t \sum_i p_i^{1-\alpha t} \\ &= Z_\alpha^{1+t} = Z_\alpha^{1/\alpha} = 2^{tH_\alpha(p)}, \end{aligned}$$

where

$$H_\alpha(p) = \frac{1}{1-\alpha} \log_2 \sum_i p_i^\alpha.$$

Consequently the exact continuous values are

$$L_t^* = H_\alpha(p)$$

and

$$T_t^* = \frac{2^t}{2^t - 1} 2^{tH_\alpha(p)} - \frac{1}{2^t - 1}.$$

This proves the proposed order $\alpha = 1/(1+t)$ and repairs the claim about the raw amortized value.

Desk-proved. The $t \rightarrow 0$ limit recovers the unit-price model: $C_0(d) = 1 + d$, $\alpha \rightarrow 1$, $d_i^* = -\log_2 p_i$, and $T_0^* = 1 + H_1(p)$. The apparent singularity in the geometric-series formula is removable.

3. What Campbell's theorem actually says

Verified. Campbell's 1965 paper defines, for a D -ary uniquely decipherable code with lengths n_i ,

$$L(t) = \frac{1}{t} \log_D \sum_i p_i D^{tn_i},$$

with the arithmetic mean as the $t \rightarrow 0$ limit and the maximum length as $t \rightarrow \infty$. It relates this exponential mean to Rényi entropy of order $(1+t)^{-1}$; block coding makes the per-symbol gap arbitrarily small. This normalization and order are stated in the publisher's record for Campbell 1965.

Desk-proved. The one-letter binary form, which is all this machine needs, is

$$H_\alpha(p) \leq L_t(d) < H_\alpha(p) + 1.$$

The lower bound holds for every integer distance assignment satisfying Kraft. Indeed,

$$\begin{aligned} Z_\alpha &= \sum_i (p_i 2^{d_i})^\alpha (2^{-d_i})^{1-\alpha} \\ &\leq \left(\sum_i p_i 2^{d_i} \right)^\alpha \left(\sum_i 2^{-d_i} \right)^{1-\alpha} \\ &\leq M_t(p, d)^\alpha \end{aligned}$$

by Hölder and Kraft. Hence $M_t \geq Z_\alpha^{1/\alpha} = 2^{tH_\alpha}$.

For achievability, take the escort shares $r_i = p_i^\alpha / Z_\alpha$ and integer lengths

$$d_i = \lceil -\log_2 r_i \rceil.$$

Then $2^{-d_i} \leq r_i$, so Kraft holds, while $d_i < -\log_2 r_i + 1$. It follows that

$$M_t(p, d) < 2^t Z_\alpha^{1/\alpha} = 2^{t(H_\alpha+1)}.$$

For blocks of length m , apply the same construction to the product distribution and divide the resulting Campbell length by m ; Rényi additivity changes the $+1$ into $+1/m$. This is the “arbitrarily close” part of Campbell’s coding theorem.

Desk-proved, comparison. Since raw travel is affine in M_t , the machine’s exact integer bounds are

$$\frac{q}{q-1} 2^{tH_\alpha} - \frac{1}{q-1} \leq T_t(p, d) < \frac{q}{q-1} 2^{t(H_\alpha+1)} - \frac{1}{q-1}.$$

Thus the two problems coincide in constraint, optimizer, and ordering of assignments. They differ in reported units and in composition: Campbell’s logarithm turns products into sums; raw machine travel does not.

4. The discrete/counting version

Let positive touch counts be $m_i \in \mathbb{N}_{>0}$ and $n = \sum_i m_i$. This matches the positivity hypothesis already used by `Adic.Entropy.empiricalDists_kraft`; zero-count heads should either be removed or handled by a separate inactive-head mounting lemma.

4.1 The honest discrete theorem still has non-natural values

Desk-proved. Substituting $p_i = m_i/n$ into the continuous optimum cancels n from the *total* moment:

$$n M_t^* = \left(\sum_i m_i^\alpha \right)^{1/\alpha}.$$

For every Kraft-feasible natural distance vector,

$$\left(\sum_i m_i^\alpha \right)^{1/\alpha} \leq \sum_i m_i 2^{t d_i},$$

and escort-ceiling distances achieve a factor less than 2^t above that benchmark. These are finite, discrete coding statements, but for $t = 1/2$ or $1/3$ their values are not naturals.

Desk-proved, repair of the proposed assignment. The bare formula

$$d_i = \lceil \alpha \log_2(n/m_i) \rceil$$

is not generally Kraft-feasible. For eight equally frequent heads and $\alpha = 2/3$, it assigns every head distance 2, giving Kraft mass $8/4 = 2$. The missing term is $\log_2 Z_\alpha$, equivalently the escort normalization. Any mechanization statement must include it, either explicitly or through a common shift computed from the preliminary masses.

4.2 Exact all-natural macro-level formulation

Desk-proved design. For rational $t = a/b$ with $a, b \geq 1$, group b binary levels into one macro-level. A head gets macro-distance $e_i \in \mathbb{N}$, corresponding to binary distance $d_i = b e_i$. Define, entirely in $\mathbb{N}$,

$$\text{MacroKraft}_b(e) : \sum_i 2^{b(E-e_i)} \leq 2^{bE}, \quad E = \max_i e_i,$$

$$A_a(m, e) = \sum_i m_i 2^{a e_i},$$

$$B_a(e) = \sum_{j=0}^e 2^{a j},$$

$$\text{Travel}_a(m, e) = \sum_i m_i B_a(e_i).$$

MacroKraft is exactly $\sum_i 2^{-b e_i} \leq 1$ after clearing a power-of-two denominator. The order selected by this pair is

$$\alpha = \frac{b}{a+b}.$$

For the Euclidean variants $a = 1, b = k$, this is $k/(k+1)$.

Desk-proved. Cumulative travel and the moment satisfy an exact natural identity, with no division:

$$(2^a - 1) \text{Travel}_a(m, e) + n = 2^a A_a(m, e).$$

This is the all-natural replacement for the affine geometric-series formula.

Define natural roots by

$$\begin{aligned} \lfloor \sqrt[r]{x} \rfloor_{\mathbb{N}} &= \max\{u : u^r \leq x\}, \\ \lceil \sqrt[r]{x} \rceil_{\mathbb{N}} &= \min\{u : x \leq u^r\}. \end{aligned}$$

Put

$$l_i = \left\lfloor \sqrt[a+b]{m_i^b} \right\rfloor_{\mathbb{N}}, \quad u_i = \left\lceil \sqrt[a+b]{m_i^b} \right\rceil_{\mathbb{N}}, \quad W = \sum_i l_i, \quad U = \sum_i u_i.$$

The floor and ceiling escort masses differ by at most one per head: $W \leq \sum_i m_i^\alpha \leq U \leq W + h$, where h is the number of heads. This is the promised linear head-count slack, expressed before any real logarithm is introduced.

Desk-proved, achievability. Define e_i as the least natural with

$$U \leq u_i 2^{be_i}.$$

Then MacroKraft holds. To see it, multiply each desired Kraft term at common width E by U , use the defining inequality pointwise, sum, and cancel the positive U .

Let

$$R = \left\lceil \sqrt[b]{U^a} \right\rceil_{\mathbb{N}}.$$

The achieved moment has the exact natural bound

$$A_a(m, e) \leq 2^a U R.$$

For each head, minimality gives $u_i 2^{be_i} < 2^b U$ (the $e_i = 0$ case is easier), while $m_i^b \leq u_i^{a+b}$. Raising the desired pointwise bound to the b th power gives

$$(m_i 2^{ae_i})^b < (2^a u_i)^b U^a \leq (2^a u_i R)^b.$$

Monotonicity of natural powers and summation finish the proof. For $a = 1, b = k$ this simplifies to

$$A_1(m, e) \leq 2U \left\lceil \sqrt[k]{U} \right\rceil_{\mathbb{N}}.$$

The factor 2^a is exactly the one-macro-digit ceiling slack; in binary Campbell-length units it is an additive b .

Desk-proved statement; wall at mechanization until the hard rung is proved. For every MacroKraft assignment, the all-natural converse is

$$W^{a+b} \leq A_a(m, e)^b.$$

A stronger denominator-cleared form is

$$W^{a+b} 2^{abE} \leq A_a(m, e)^b \left(\sum_i 2^{b(E-e_i)} \right)^a.$$

This is rational Hölder with every denominator and fractional exponent cleared. MacroKraft reduces the rightmost factor to 2^{abE} and cancellation gives the first inequality. The statement is finite and purely natural; the missing artifact is a Lean proof of this generalized Hölder polynomial inequality. This is the hard rung, not a hidden use of real analysis.

4.3 Relating the macro model to literal $2^{\ell/k}$ prices

Desk-proved design. Macro-levels preserve the exponent and Rényi order but coarsen binary distances. For a faithful natural-valued binary schedule, define

$$\hat{c}_k(\ell) = \left\lceil \sqrt[k]{2^\ell} \right\rceil_{\mathbb{N}}, \quad \hat{C}_k(d) = \sum_{\ell=0}^d \hat{c}_k(\ell).$$

This is all-natural and satisfies

$$2^{\ell/k} \leq \hat{c}_k(\ell) < 2^{\ell/k} + 1 \leq 2 \cdot 2^{\ell/k}.$$

Consequently literal algebraic travel and rounded-natural travel differ by at most $d + 1$ per touch and by at most a factor two. Grouping levels into blocks of k then compares $\hat{C}_k$ with the macro schedule within a constant depending only on k .

Recommendation (design, not decision). Mechanize the macro theorem first because every statement above is in the existing Nat/power-of-two style. Then add the rounded binary schedule and constant-factor sandwich. Use Real or exact algebraic numbers only if the paper truly needs the literal constants $\sqrt{2}$ and $\sqrt[3]{2}$; the complexity exponents and Rényi orders do not need them.

5. The two Euclidean instantiations

Let

$$Z_\alpha = \sum_i p_i^\alpha, \quad q_k = 2^{1/k}.$$

Desk-proved. The exact continuous table is:

ambient dial	t	α	optimal share $2^{-d_i^*}$	optimal moment	optimal cu- mulative travel
$k = 2$	$1/2$	$2/3$	$p_i^{2/3}/Z_{2/3}$	$Z_{2/3}^{3/2} = 2^{H_{2/3}/2}$	$\frac{\sqrt{2}}{\sqrt{2}-1} 2^{H_{2/3}/2} - \frac{1}{\sqrt{2}-1}$
$k = 3$	$1/3$	$3/4$	$p_i^{3/4}/Z_{3/4}$	$Z_{3/4}^{4/3} = 2^{H_{3/4}/3}$	$\frac{\sqrt[3]{2}}{\sqrt[3]{2}-1} 2^{H_{3/4}/3} - \frac{1}{\sqrt[3]{2}-1}$

Worked three-head sanity check

Take

$$p = (1/2, 1/4, 1/4).$$

Desk-proved baseline. For the unit-price model, Shannon's assignment is already integral: $d = (1, 2, 2)$, Kraft is tight, $H_1 = 3/2$, and expected travel $\sum p_i(1 + d_i) = 5/2$.

Desk-proved, $k = 2$. Here

$$Z_{2/3} = 2^{-2/3} + 2 \cdot 2^{-4/3} \approx 1.423661, \quad H_{2/3} \approx 1.528817.$$

The escort shares and real distances are

$$r \approx (0.442493, 0.278753, 0.278753), \quad d^* \approx (1.176272, 1.842939, 1.842939).$$

Kraft sums to one by construction. Direct substitution gives

$$M_{1/2}^* \approx 1.698673, \quad T_{1/2}^* \approx 3.385420.$$

Rounding with the escort ceiling again gives $d = (2, 2, 2)$, but that is not the best integer code. The feasible code $(1, 2, 2)$ has

$$M_{1/2} = \frac{1}{2}\sqrt{2} + \frac{1}{2} \cdot 2 \approx 1.707107, \quad T_{1/2} = 2 + \sqrt{2} \approx 3.414214,$$

and Campbell length $2 \log_2 M_{1/2} \approx 1.543107$, between $H_{2/3}$ and $H_{2/3} + 1$.

Desk-proved, $k = 3$. Here

$$Z_{3/4} = 2^{-3/4} + 2 \cdot 2^{-3/2} \approx 1.301710, \quad H_{3/4} \approx 1.521634,$$

with

$$r \approx (0.456786, 0.271607, 0.271607), \quad d^* \approx (1.130408, 1.880408, 1.880408).$$

Direct substitution gives

$$M_{1/3}^* \approx 1.421300, \quad T_{1/3}^* \approx 3.042178.$$

For the integer code $(1, 2, 2)$,

$$M_{1/3} = \frac{1}{2}2^{1/3} + \frac{1}{2}2^{2/3} \approx 1.423661, \quad T_{1/3} \approx 3.053622,$$

and $3 \log_2 M_{1/3} \approx 1.528817$, again above the corresponding Rényi entropy.

Desk-proved, interpretation. The rare heads move from ideal Shannon distance 2 to about 1.84 or 1.88, while the hot head moves from 1 to about 1.18 or 1.13. The assignment flattens: expensive depth pulls rare heads inward and gives up some of the hot head's privileged nearness. Integer prefix geometry can hide that motion in small examples; the real relaxation exposes the direction.

6. Expo 2 at k dimensions

Verified baseline. Expo 2 has already mechanized that Kraft is equivalent to an antichain mounting and that unit-price touch cost is bounded above and below by the Shannon count form, with explicit natural slack. None of those mounting facts depends on a level-price schedule.

Desk-proved exposition sketch. The Euclidean continuation can therefore begin without changing a single mounting sentence:

Three heads still occupy disjoint dyadic shares. The same Kraft inequality says which shares fit. What changes is how far-away shares are charged. At k dimensions, a level- ℓ crossing costs $2^{\ell/k}$, so cumulative travel to distance d is geometric. The best shares are no longer p_i but the escort shares $p_i^{k/(k+1)} / \sum_j p_j^{k/(k+1)}$. Rare heads are pushed less deep because depth is now superlinearly priced. The normalized optimum is Rényi entropy of order $k/(k+1)$; raw travel is exponential in it.

Desk-proved. The before/after table for the exposition is:

feature	informational machine, $c \equiv 1$	k -dimensional price, $c(\ell) = 2^{\ell/k}$
mounting law	$\sum_i 2^{-d_i} \leq 1$	unchanged
ideal share	p_i	$p_i^{k/(k+1)} / Z$
ideal distance	$\log_2(1/p_i)$	$\log_2 Z + \frac{k}{k+1} \log_2(1/p_i)$
optimized normalized value	$H_1(p)$	$H_{k/(k+1)}(p)$
optimized raw touch travel	$1 + H_1(p)$	$\frac{q_k}{q_k-1} 2^{H_{k/(k+1)}/k} - \frac{1}{q_k-1}$
integer rounding	Shannon/Huffman	Campbell/generalized Huffman

Desk-proved worked example. The three-head vector $(1/2, 1/4, 1/4)$ from Section 5 is a useful successor to [59]. It keeps Kraft tight at integer distances $(1, 2, 2)$ while the relaxed distances visibly flatten. [59]'s uniform vector $(1/3, 1/3, 1/3)$ remains uniform under every Rényi order

and still uses integer distances $(2, 2, 2)$; that makes it a good mounting example but a bad example of the changed optimizer.

The corrected one-line slogan for the page is:

Same Kraft geometry, new travel objective: Shannon at unit prices, Rényi after exponential normalization.

7. Which schedules are well behaved?

Let $c : \mathbb{N} \rightarrow \mathbb{R}_{\geq 0}$ be a nonnegative monotone level price and

$$C(d) = \sum_{\ell=0}^d c(\ell).$$

For a grade- n tree, write $N = 2^n$.

7.1 The streaming boundary

Desk-proved, correction of the literal counting claim. In a cyclic odometer sweep of all N leaves, the number of paid crossings at scale ℓ is

$$2^{n-\ell}, \quad 0 \leq \ell \leq n,$$

in aggregate. It is not the crossing count of each fixed edge. A one-way sweep differs by at most one crossing at each scale; the cyclic form is the clean exact identity. Its priced movement is

$$S_n(c) = \sum_{\ell=0}^n c(\ell) 2^{n-\ell} = N \sum_{\ell=0}^n c(\ell) 2^{-\ell}.$$

Leaf reads add another $\Theta(N)$.

Desk-proved. Because the terms are nonnegative, uniformly $O(1)$ amortized movement per leaf holds if and only if

$$\sum_{\ell \geq 0} c(\ell) 2^{-\ell} < \infty.$$

No doubling hypothesis is needed for this equivalence.

Desk-proved. For $c_s(\ell) = 2^{\ell/s}$:

range	one random access, $C_s(n)$	full sweep, $S_n(c_s)$
$s > 1$	$\Theta(N^{1/s})$	$\Theta(N)$
$s = 1$	$\Theta(N)$	$\Theta(N \log N)$
$0 < s < 1$	$\Theta(N^{1/s})$	$\Theta(N^{1/s})$

Desk-proved, correction. The $s = 1$ point has tape-like random access but not tape-like streaming. A Turing tape reaches an arbitrary cell in $\Theta(N)$ and scans N adjacent cells in $\Theta(N)$; the priced dyadic path pays one $\Theta(N)$ contribution at every scale, hence the extra logarithm. So the proposed dial is exact for random access,

$$s = 1 \text{ linear}, \quad s = 2, 3 \text{ Euclidean}, \quad s \rightarrow \infty \text{ informational/logarithmic},$$

but “ $s = 1$ is the Turing tape” is false as a whole-machine statement. No schedule with $c(\ell) = \Theta(2^\ell)$ can also satisfy the streaming summability condition.

Desk-proved. Near-tape schedules show the sharp boundary. If $c(\ell) = 2^\ell/(\ell + 1)^r$, then random access is $\Theta(N/(\log N)^r)$, while streaming is linear exactly when $r > 1$; $r = 1$ gives $\Theta(N \log \log N)$ for the sweep.

7.2 Growth map

Desk-proved. The useful classification is by both $C(n)$ and the weighted series above; neither alone controls the other.

- If c is bounded above and bounded away from zero, then $C(n) = \Theta(n)$: random access is $\Theta(\log N)$ and streaming is linear. This is the informational/hyperbolic fiber.
- If $c(\ell) = \Theta(\ell^r)$, then $C(n) = \Theta(n^{r+1})$: random access is polylogarithmic in N and streaming remains linear. Polynomial growth of C in *tree depth* is therefore not Euclidean; the corresponding ball growth is stretched exponential. Euclidean polynomial volume growth requires $C(n)$ exponential in depth.
- If c is subexponential, for example $2^{\sqrt{\ell}}$, random access is subpolynomial in N and streaming is still linear.
- If $c(\ell) = \Theta(2^{\ell/s})$ with $s > 1$, random access is $\Theta(N^{1/s})$ and streaming is linear. Integer s gives the Euclidean dimensions; noninteger s is the fractal dial.
- At the critical envelope 2^ℓ the slowly varying factor decides streaming, as above.
- Faster-than-critical exponential and super-exponential schedules destroy constant-amortized streaming. When the last term dominates, random access and a full sweep can have the same order: the rare top boundary costs as much as the whole traversal.

7.3 What a doubling axiom buys

Desk-proved. A cumulative doubling bound

$$C(d + 1) \leq KC(d)$$

is a useful *rounding-stability* axiom: increasing a real optimum by one integer level loses at most a factor K . It also makes local changes of distance quantitatively tame. It is not the general well-behavedness axiom:

- streaming needs the independent summability condition;
- exact random-access cost is $C(n)$ and needs no doubling;
- Rényi identification needs $C(d)$ to be affine in an exponential, or at least two-sided comparable to one;

- doubling alone neither supplies additivity nor selects a Rényi order.

A clean theorem interface should therefore expose separate hypotheses: nonnegative monotonicity for basic cost lemmas, summability for streaming, cumulative doubling for integer rounding, and two-sided geometric growth for Campbell/Rényi comparisons. Conflating these into one “reasonable schedule” class would hide real boundary cases.

8. Why exponentials are canonical—and the precise limit of that claim

Verified. Rényi’s original 1961 characterization starts from a Kolmogorov–Nagumo quasi-arithmetic mean and an additivity requirement for independent experiments. Under its continuity and monotonicity postulates, the admissible mean generator is linear or exponential; the corresponding entropies are Shannon and the Rényi family. See Theorems 2 and 3 in Rényi, “On Measures of Entropy and Information”.

Desk-proved in the present setting. For independent head groups with probabilities p_i and s_j , and additive product distance $d_i + e_j$,

$$\sum_{i,j} p_i s_j 2^{t(d_i + e_j)} = \left(\sum_i p_i 2^{t d_i} \right) \left(\sum_j s_j 2^{t e_j} \right).$$

Therefore $L_t = (1/t) \log_2 M_t$ is additive. Its continuous optimum is Rényi entropy, which is itself additive on product distributions. The $t \rightarrow 0$ limit is the arithmetic mean and Shannon entropy.

Desk-proved, qualification. Raw cumulative travel is not additive: it is an affine function of M_t . It becomes additive only after the normalization

$$\Phi_t(T) = \frac{1}{t} \log_2 \left(\frac{(2^t - 1)T + 1}{2^t} \right) = L_t.$$

Likewise, integer code optima are additive only up to rounding slack. The axiomatic claim is therefore not “all additive machine costs are exponential.” The honest statement is:

Among continuous, strictly monotone quasi-arithmetic aggregations of additive distances that compose over independent products, the generators are linear or exponential. Their coding optima are Shannon or Rényi. Geometric level schedules are exactly the machine schedules whose cumulative cost is affine in the exponential generator.

Conjectured extension. A machine-native characterization could take product composition, translation covariance under a common distance shift, and continuity/monotonicity as axioms and derive the same linear/exponential dichotomy. The finite functional-equation proof is not supplied here. Rényi’s theorem strongly suggests it, but citing the information-measure theorem is not itself a proof for arbitrary machine schedule functionals.

9. Lean mechanization ladder

Recommendation (design, not decision). The module name should be `Adic.Campbell`. Use `h` for the number of heads and reserve `k` for ambient dimension.

Rung 1: macro Kraft arithmetic

Desk-proved design, easy. Define:

```
blockKraftMassAt (b width : Nat) (dists : Fin h → Nat) : Nat
BlockKraft0k (b : Nat) (dists : Fin h → Nat) : Prop
expMomentCost (a : Nat) (counts dists : Fin h → Nat) : Nat
blockTravel (a dist : Nat) : Nat
blockTravelCost (a : Nat) (counts dists : Fin h → Nat) : Nat
```

Prove:

```
blockKraft0k_iff
blockKraft_shift
blockTravel_geometric
blockTravelCost_moment_identity
```

The key exact statement is

```
(2 ^ a - 1) * blockTravelCost a counts dists + finSum counts =
  2 ^ a * expMomentCost a counts dists
under 0 < a.
```

Rung 2: integer roots

Desk-proved design, routine but library-sensitive. Either reuse Mathlib's natural root API or define least roots locally. Prove:

```
floorRoot_pow_le
lt_succ_floorRoot_pow
le_ceilRoot_pow
ceilRoot_le_floorRoot_add_one
pow_le_pow_iff
```

with positive exponents. Friction risk: names and available generic Nat root lemmas may dictate the local API.

Rung 3: escort weights and distances

Desk-proved design, medium. Define:

```
escortFloor (a b count : Nat) := floorRoot (a + b) (count ^ b)
escortCeil (a b count : Nat) := ceilRoot (a + b) (count ^ b)
escortMass (a b : Nat) (counts : Fin h → Nat) : Nat
blockInverseFrequencyDist (b total share : Nat) : Nat
campbellDists (a b : Nat) (counts : Fin h → Nat) : Fin h → Nat
```

where `blockInverseFrequencyDist` is the least e satisfying $\text{total} \leq \text{share} * 2 ^ (b * e)$. Prove:

```
blockInverseFrequencyDist_spec
blockInverseFrequencyDist_minimal
campbellDists_blockKraft
escortMass_floor_ceiling_slack
```

Hypotheses: $0 < a$, $0 < b$, and every count positive.

Rung 4: achievability

Desk-proved design, medium-hard. With $U := \text{finSum } (\text{escortCeil } a \ b \circ \text{counts})$ and $R := \text{ceilRoot } b \ (U \wedge a)$, prove:

```
expMomentCost_campbell_upper :  
  expMomentCost a counts (campbellDists a b counts) ≤  
    2 ^ a * U * R
```

Then transfer it through `blockTravelCost_moment_identity`. This rung is pointwise power arithmetic plus a finite sum; it should not require analysis.

Rung 5: cleared Hölder converse

Wall / hard rung. Prove a reusable natural inequality:

```
campbell_holder_cleared :  
  W ^ (a + b) * 2 ^ (a * b * E) ≤  
    expMomentCost a counts dists ^ b *  
    blockKraftMassAt b E dists ^ a
```

where

```
W = finSum (fun i => floorRoot (a + b) (counts i ^ b))  
E = maxDist dists
```

and the usual positivity hypotheses hold. Then derive:

```
expMomentCost_campbell_lower  
blockTravelCost_campbell_lower
```

from `BlockKraft0k`. This is the exact all-natural Campbell converse and the only genuinely difficult rung. A multinomial/counting proof is preferred if manageable; a first implementation through nonnegative rationals followed by denominator clearing would still keep the public theorem all-natural, but would be a larger dependency than `Entropy.lean` currently needs.

Rung 6: $k = 2, 3$ binary schedules

Desk-proved design, medium. Define:

```
natLevelPrice (k level : Nat) := ceilRoot k (2 ^ level)  
natTravel (k dist : Nat) := ∑ level in Finset.range (dist + 1),  
  natLevelPrice k level
```

Prove the block sandwich for $k=2$ and $k=3$, then expose named instances:

```
campbellOrder2_num = 2  
campbellOrder2_den = 3  
campbellOrder3_num = 3  
campbellOrder3_den = 4  
natTravel_two_dimensional_bounds  
natTravel_three_dimensional_bounds
```

No theorem should claim literal equality with $2^{\ell/k}$ while its codomain is `Nat`.

10. Stretch: the WBE rhyme

Verified. West, Brown, and Enquist derive quarter-power scaling from a space-filling branching transport network with size-invariant terminal units and an energy-dissipation optimization; the original model is West–Brown–Enquist 1997. Their later dimensional summary writes the exponent as $d/(d + 1)$ (West–Brown 2005), which is $3/4$ in three dimensions. The model and the empirical universality of the exact $3/4$ exponent have been disputed; see, for example, Dodds–Rothman–Weitz 2001.

Desk-proved negative result. The WBE optimization is not a Campbell problem after an evident change of variables. Campbell has a probability vector, a Kraft/prefix-packing constraint, and minimizes an exponential mean of code lengths. WBE constrains a space-filling vascular network, terminal units, fluid flow, and hydrodynamic energy. There is no symbol probability distribution, no prefix budget, and no exponential mean whose logarithm is an information measure. The two problems do share homogeneous branching and a Lagrange balance between d geometric directions and one additional constraint, so the ratio $d/(d + 1)$ can appear in both.

Conclusion on the stretch. At present the coincidence is a useful rhyme, not an identification. Claiming a Campbell theorem in disguise would require an explicit map carrying WBE’s feasible networks to Kraft-feasible shares and its dissipation objective to an exponential moment. No such map is visible, and the empirical/theoretical debate around WBE makes it a poor foundation for a machine theorem. Keep it as a one-paragraph cross-disciplinary observation, not as evidence.

11. Open gaps and requested repairs

Desk-proved summary. The main proposed correspondence survives with these repairs:

1. include the escort normalization $\log_2 Z_\alpha$ in the optimal distance;
2. say raw travel is affine in 2^{tH_α} , while Campbell’s normalized logarithm equals H_α ;
3. keep Kraft geometry unchanged;
4. qualify $s = 1$ as a random-access tape endpoint only;
5. distinguish polynomial growth in depth from Euclidean polynomial ball growth;
6. choose explicitly how irrational Euclidean prices enter an all-natural mechanization.

Wall report. No mathematical contradiction was found in the key order $\alpha = 1/(1 + t)$, and no sub-question consumed the track without a result. The honest unresolved proof obligation is the natural cleared-Hölder rung. The honest design wall is exact irrational pricing with a Nat codomain. Both are isolated above; neither blocks the desk result or the recommended macro-level theorem.

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